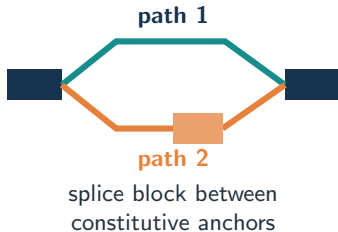


Differential splicing from equivalence-class counts

Local splice-path effects propagated through isoforms and
primer-aware observations

The target is a local splice choice with biological replication



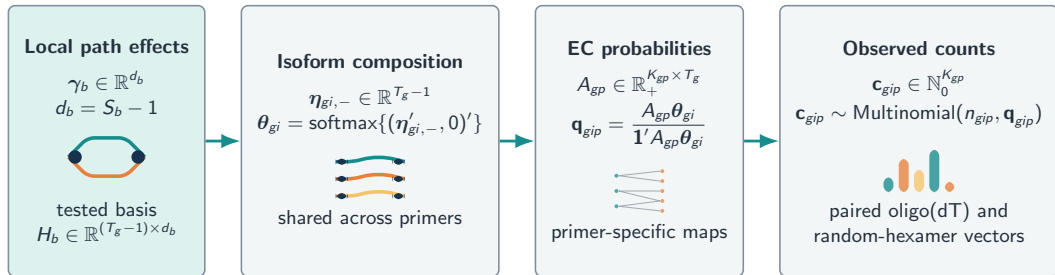
Two questions

Cell type, does path usage differ across cell types while condition is unrestricted?

Condition, does path usage differ across conditions within one cell type?

- ▶ One hypothesis per nonredundant local path partition
- ▶ Mouse random effects represent biological replication
- ▶ Cells contribute to mouse $\times$ condition $\times$ cell-type pseudobulks, never independent replicates

Forward model, splice-path effects $\rightarrow$ isoforms $\rightarrow$ ECs $\rightarrow$ counts



Ambiguous ECs remain ambiguous, primer design enters through A_{gp} , and the biological effect is tested only in the local path subspace.

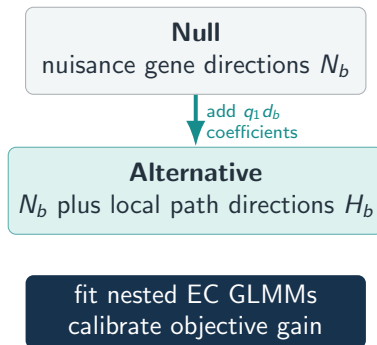
The nested test isolates the block contrast without fixing the rest of the gene

Isoform-logit predictor

The tested block basis $H_b \in \mathbb{R}^{(T_g-1) \times d_b}$ lifts an orthonormal path contrast into isoform space. The nuisance basis $N_b \in \mathbb{R}^{(T_g-1) \times (T_g-S_b)}$ spans every orthogonal gene-level direction.

The null predictor is $\mathcal{F}_{bi}^{(0)} = B'_0 Z'_{bi} + \sum_j W_{bij} N_b \alpha_{bj}$, while the alternative adds $\sum_j W_{bij} H_b \gamma_{bj}$.

- ▶ $Z_b \in \mathbb{R}^{l_b \times q_0}$ carries nuisance covariates
- ▶ $W_b \in \mathbb{R}^{l_b \times q_1}$ carries the tested cell-type or condition design
- ▶ The alternative adds $q_1 d_b$ local path coefficients, but remains unrestricted in free isoform-logit space



Calibration matters

The ordinary multilevel chi-square LRT is anti-conservative under structured nulls. The primary mouse result uses simulated nulls, cross-fitted tail modeling, leave-block-out calibration, and BH.

Two brain datasets, one subject-level analysis design

Mouse

GSE189033

Biology, 5xFAD amyloidosis crossed with Csf1r Δ FIRE microglial deficiency, including PBS and microglia-transplant arms

Samples, 48 mice, six groups, balanced 4 female and 4 male per group, age 5–6 months

Assay, SPLiT-seq single-nucleus RNA-seq, paired oligo(dT) and random-hexamer observations

Analysis strata, 936 mouse $\times$ condition $\times$ cell-type pseudobulks before feature-level support filtering

Human

GSE233208

Biology, cortical postmortem tissue spanning control, sporadic Alzheimer disease, Down syndrome, and Down syndrome with AD

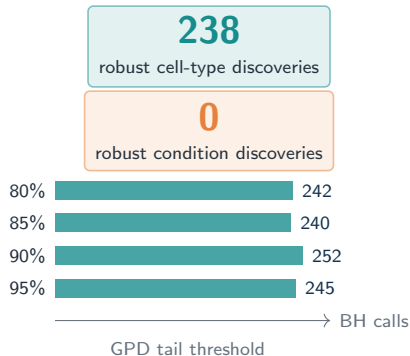
Scale, 396,277 annotated nuclei represented by 40 paired sequencing runs in five batches

Assay, Parse Biosciences Evercode WT v1 single-nucleus RNA-seq

Role, independent cross-species sensitivity analysis with the same EC GLMM fitting procedure

Sources, Spangenberg et al., Cell Reports 2022, GEO GSE189033; Miyoshi et al., Nature Genetics 2024, GEO GSE233208. Counts shown are the analysis metadata summarized in this repository.

Calibrated mouse analysis finds cell-type splicing, not robust condition effects



- ▶ 4,825 cell-type path partitions and 14,833 condition-within-cell-type tests
- ▶ Aggregate null rejection matches 0.05, 0.01, and 0.001 after cross-fitted calibration
- ▶ A robust event passes BH across every 80–95% tail threshold
- ▶ The primary logistic-normal likelihood allows pseudobulk extra-multinomial variation shared across the two primer halves

Interpretation

Cell identity is associated with many local isoform choices. No condition effect survives the prespecified calibration sensitivity analysis.

Matched comparator yields span different feature universes

Method	Unit	Mouse			Human		
		CT tests	CT BH	CT rate	CT tests	CT BH	CT rate
Tealeaf	EC block	4,825	322	6.67%	11,597	1,463	12.62%
LeafCutter	intron cluster	1,020	229	22.45%	2,155	238	11.04%
MAJIQ Heterogen	LSV edge	46	27	58.70%	582	342	58.76%
scQuint	intron group	16,081	828	5.15%	27,097	2,239	8.26%

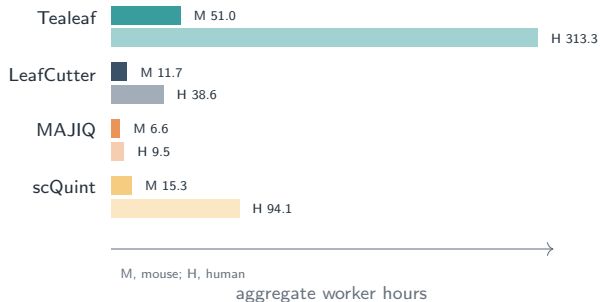
Condition



Read as descriptive yield, not a power ranking

Methods test different units and apply different filters. External cell-type results combine pairwise tests by Simes, whereas Tealeaf fits an omnibus model. The ordinary multilevel Tealeaf LRT used here is not structured-null calibrated, so these BH counts are descriptive; the 238-call calibrated mouse analysis is the validated result.

EC ambiguity and latent-Gaussian fitting carry a compute cost



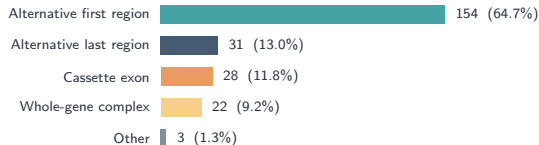
3.3×

slower than scQuint in both broad campaigns

- ▶ Same scalar Laplace LRT and analytic EC derivatives in both species
- ▶ Human has 4.3× more tests
- ▶ Median human test has 5 isoforms, 26 ECs, and 5 paths; mouse has 3, 9, and 2
- ▶ Peak memory, 4.81 GiB mouse and 8.25 GiB human

Scope starts from prepared method inputs and includes cell-type plus condition fitting and result construction; shared read processing is excluded.

Robust mouse calls are enriched for transcript-end differences



Functional precedents among robust calls

- ▶ *Ntrk2*, alternative last region separating short kinase-deficient and long TrkB paths; TrkB.T1 has a characterized astrocyte role
- ▶ *Grin1*, C1 cassette exon linked to receptor surface trafficking
- ▶ *App*, exon 15 cassette linked to polarized secretion and amyloid processing
- ▶ *Enah*, exon 11a cassette linked to actin dynamics outside brain

Boundary of the annotation

A terminal-region label means supported paths differ between a constitutive anchor and a transcript boundary. It does not by itself distinguish promoter choice, terminal exon choice, or polyadenylation.

Takeaways

Model the observations that were measured. Local path effects act through full isoforms, primer-specific EC maps, and observed pseudobulk counts without hard EC allocation.

Separate biology from calibration. The calibrated mouse analysis supports 238 robust cell-type events and no robust condition events; human ordinary-LRT yields remain sensitivity results.

Comparator yield is not accuracy. Tealeaf, LeafCutter, MAJIQ, and scQuint use different feature universes, filters, and omnibus constructions.

The remaining tradeoff is compute. Retaining EC ambiguity and repeatedly solving nested latent-Gaussian models costs about $3.3\times$ scQuint's broad worker time after analytic derivative speedups.